

MASHRUBA TASNEM OISHI

Computer Science & Engineering Graduate | AI/ML | Software Development | Education



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 PROFESSIONAL PROFILE

Computer Science & Engineering graduate (CGPA 3.64/4.00) with experience across **AI/ML research, software development, and computer science education**. Experienced in developing research-driven machine learning solutions, AI-integrated full-stack applications, and student-focused learning materials. Strong technical foundation in **Python, software development, databases, machine learning, and web technologies**, with hands-on experience in research methodology, model explainability, and application development. Demonstrated ability to apply technical knowledge to both **academic research and practical software projects**.

 CORE SKILLS

Programming Languages Python, C, C++, Java, C#, PHP, SQL	AI / Machine Learning Scikit-learn, XGBoost, Random Forest, SHAP, LIME, Pandas, NumPy	Web / Database HTML, CSS, JavaScript, Bootstrap, jQuery, PHP, MySQL, SQL Server, SSLCommerz	Tools Git, GitHub, VS Code, Visual Studio, Google Colab, Kaggle
Professional Skills Communication, Teaching & Mentoring, Analytical Thinking, Problem Solving, Teamwork			

 PROFESSIONAL EXPERIENCE

Web Developer Intern, SoftDeft Academic Internship	02/2026 – 05/2026
- Designed and developed a full-stack Animal Rescue & Sanctuary Management System using PHP, MySQL, Bootstrap, and JavaScript, featuring 4 role-based portals, 20+ relational database tables, and secure CRUD operations. - Implemented role-based authentication and authorization, input validation and sanitization, password hashing, and SQL injection prevention to strengthen application security. - Integrated SSLCommerz for secure donation and transaction processing, with appropriate validation and access controls. - Built AI-assisted and location-based features, including a Groq-powered SOS triage system, AI chatbot, and OpenStreetMap-based clinic finder, supporting faster rescue response and accessibility.	
Academic Tutor, Independent / Private Tutoring	01/2022 – Present
- Teach ICT, Computer Science, programming, Mathematics, Physics, Chemistry, Biology, and General Science across multiple academic levels. - Develop lesson plans, assignments, quizzes, and study materials tailored to individual learning needs. - Provide individualized academic guidance to strengthen conceptual understanding, problem-solving skills, and examination readiness.	

 RESEARCH & PUBLICATION

XAI-NIDS: Explainable ML Framework for Binary Network Intrusion Detection, Undergraduate Thesis	05/2025 – 05/2026
- Designed and implemented a leakage-controlled network intrusion detection framework on UNSW-NB15, integrating training-only preprocessing, fold-local oversampling, feature engineering, stratified cross-validation, and five tree-based model configurations. - Achieved 90.52% accuracy, 90.43% weighted F1, and 98.37% ROC-AUC with XGBoost on the official held-out test set. - Applied SHAP and LIME for global and local model explainability and conducted controlled feature-ablation experiments to validate feature importance. - Performed Holm-corrected McNemar testing and Nadeau–Bengio corrected t-tests to statistically validate model differences; 9 of 10 pairwise comparisons were significant on UNSW-NB15. - Independently reproduced the framework on CIC-IDS-2017, achieving 99.92% accuracy and F1 with optimized XGBoost, demonstrating cross-benchmark reproducibility. Manuscript prepared for submission.	

 EDUCATION

International University of Business Agriculture & Technology (IUBAT), B.Sc. in Computer Science & Engineering (BCSE) CGPA: 3.64 / 4.00	2022 – 2026
RAJUK Uttara Model College, Higher Secondary Certificate (HSC) GPA: 5.00 / 5.00	2017 – 2019
RAJUK Uttara Model College, Secondary School Certificate (SSC) GPA: 5.00 / 5.00	2015 – 2017

 TEACHING & RESEARCH AREAS

Teaching: Programming Fundamentals, Object-Oriented Programming, Database Management Systems, Computer Networks, Software Engineering, Management Information Systems, System Analysis & Design, Digital Logic Design, introductory Machine Learning.
Research: Artificial Intelligence, Explainable AI, Machine Learning, Trustworthy AI, Cybersecurity, Network Intrusion Detection, Data Science.

 LANGUAGES

Bengali Native	English Fluent
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